

# NuSTAR Background

Origins, Mitigations, and Modeling

Brian Grefenstette, November 11, 2024

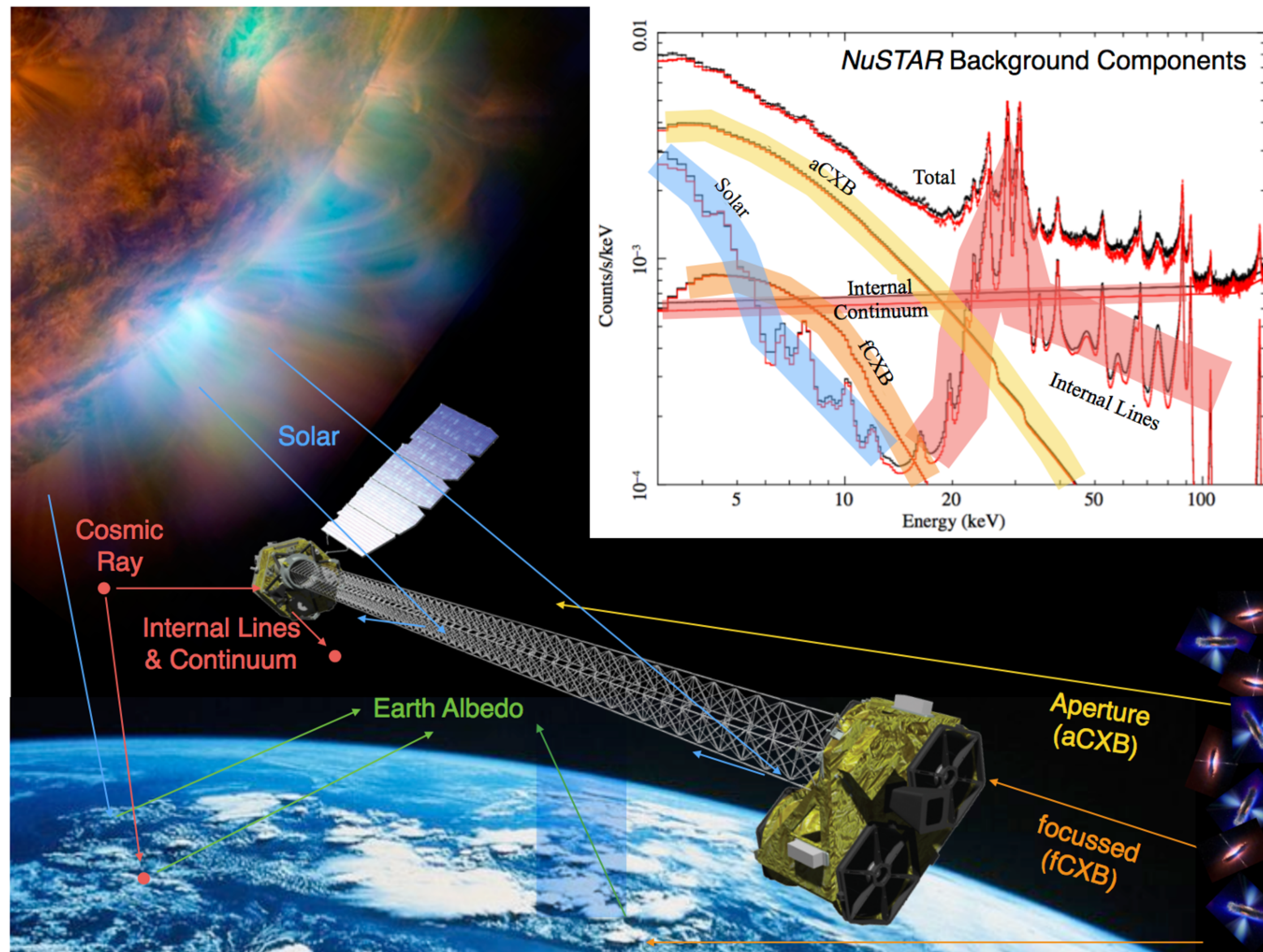
# Scope

## **How to handle and produce background models**

- Origin of the NuSTAR background
- Mitigation strategies
- Background modeling techniques



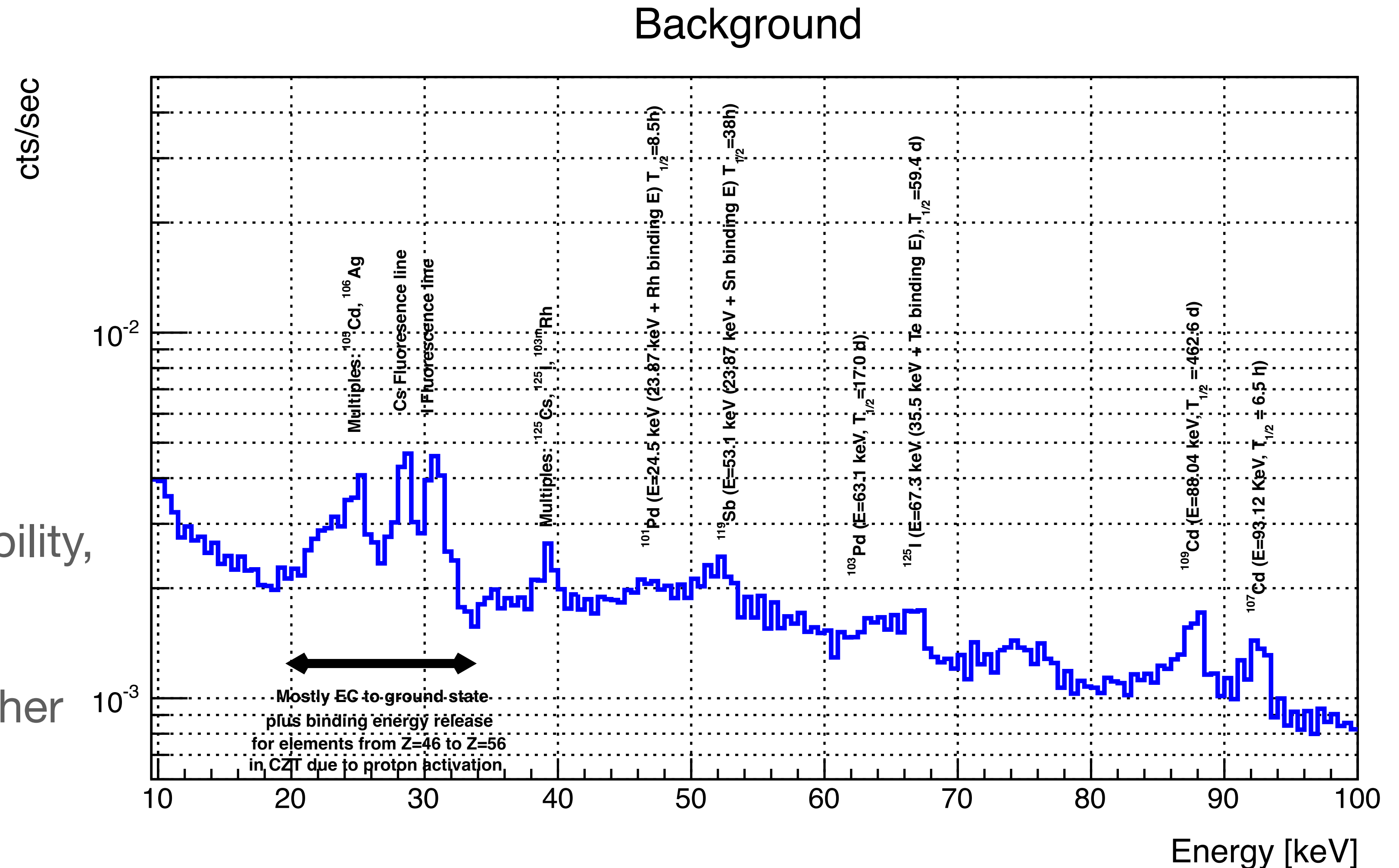
# Origin of the NuSTAR Background





# Origin of the NuSTAR Background

- Lines result (mostly) from proton activation
- Half-life ranges from hours to years
- Short-lived lines  $\rightarrow$  temporal variability, or “radiation belt memory”
- Continuum depends on space weather





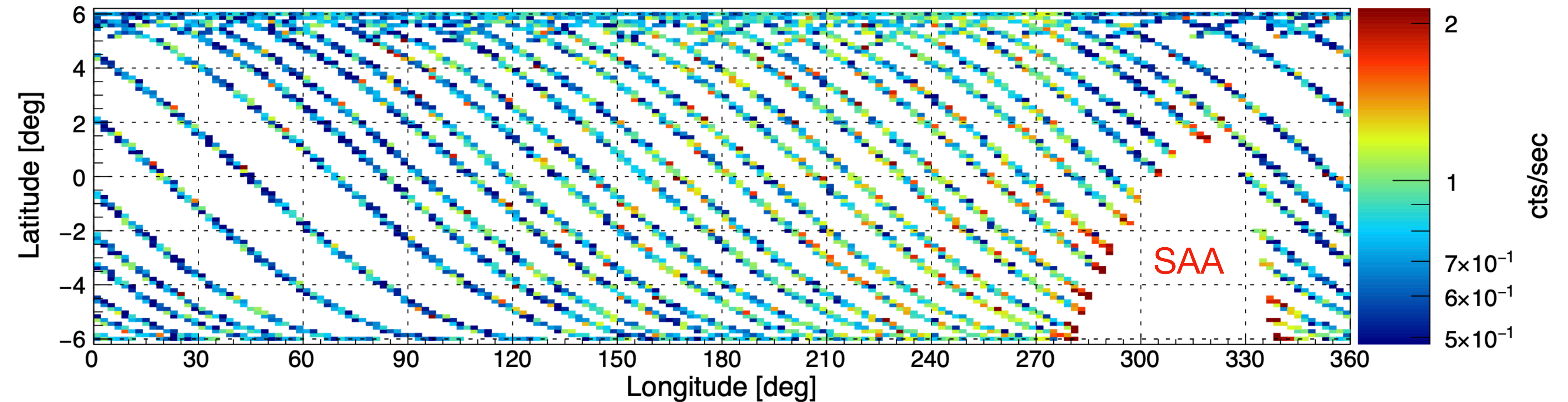
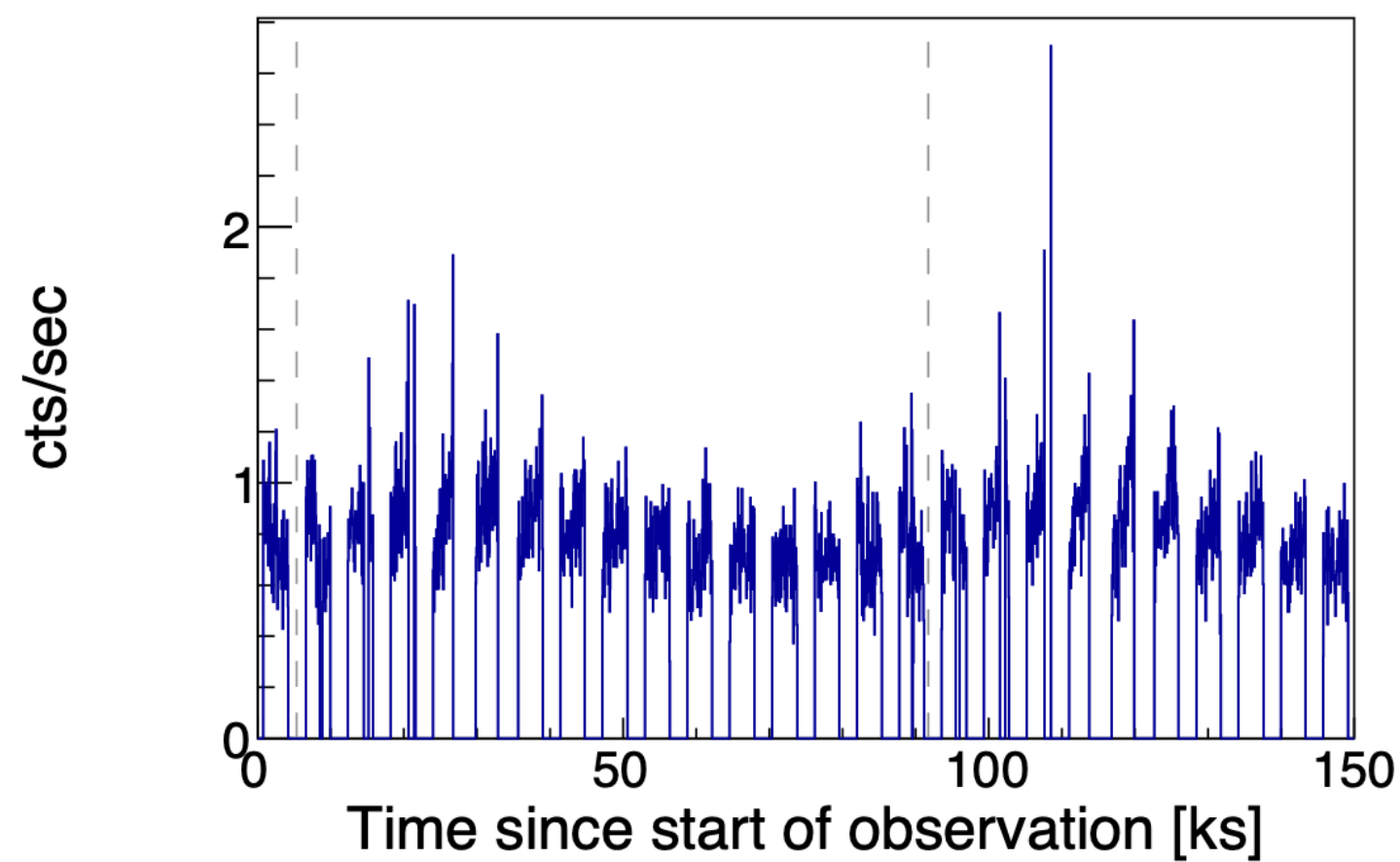
# Mitigation

## How do I know if the space weather was bad and what can I do?

- NuSTAR SOC provides automated checks and summary reports for standard filtering in nustardas
- Link here: [https://nustarsoc.caltech.edu/NuSTAR\\_Public/NuSTAROperationSite/SAA\\_Filtering/SAA\\_Filter.php](https://nustarsoc.caltech.edu/NuSTAR_Public/NuSTAROperationSite/SAA_Filtering/SAA_Filter.php)
- Available for every observation throughout the mission (updated automatically)
- Up to user to decide which combination of filters is appropriate / best utilizes the data

# Mitigation

What does a “good” background look like?



“Spikes” associated with SAA ingress/egress

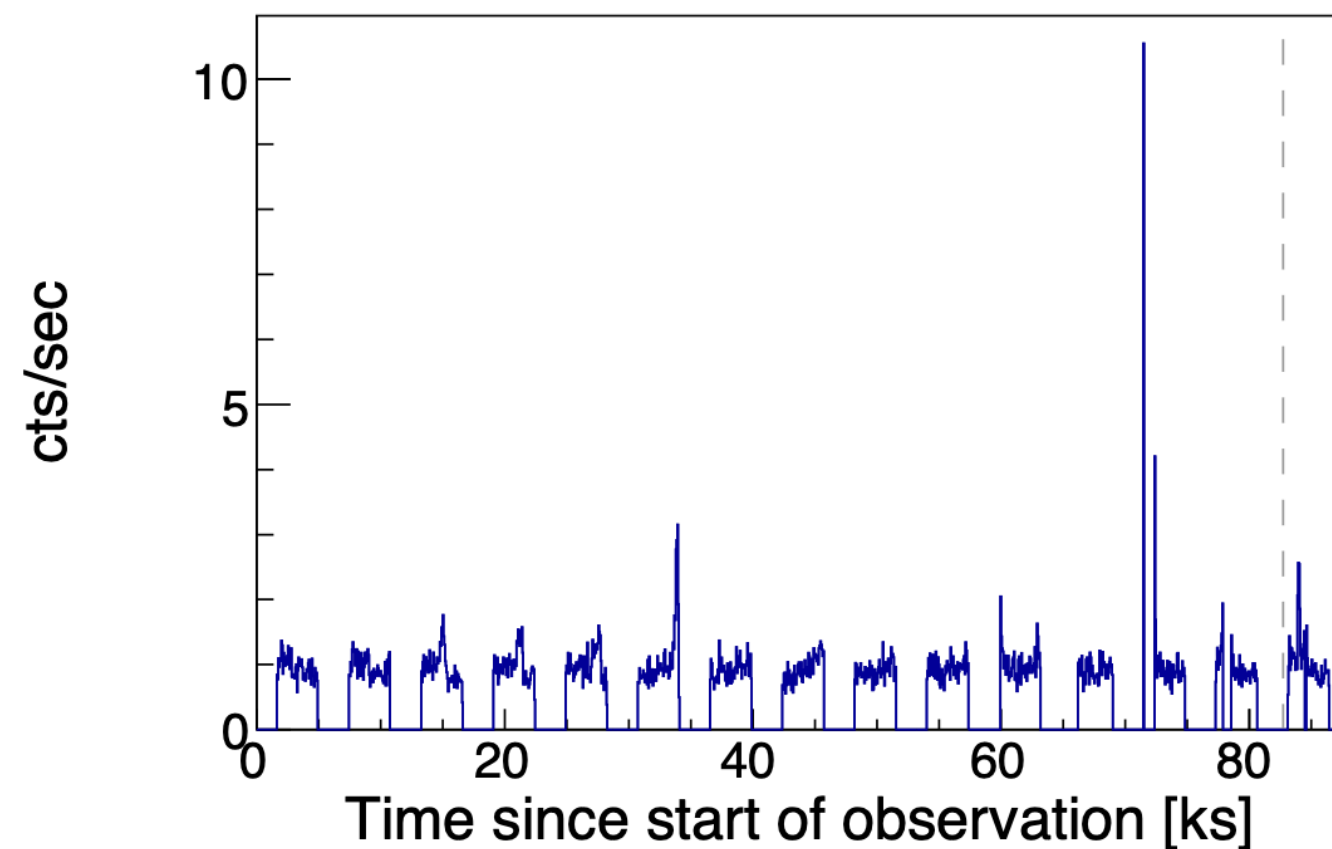
Slow variations associated with geomagnetic rigidity cutoff and orbital precession

Grefenstette et al (2022), in prep to discuss “normal” background variations

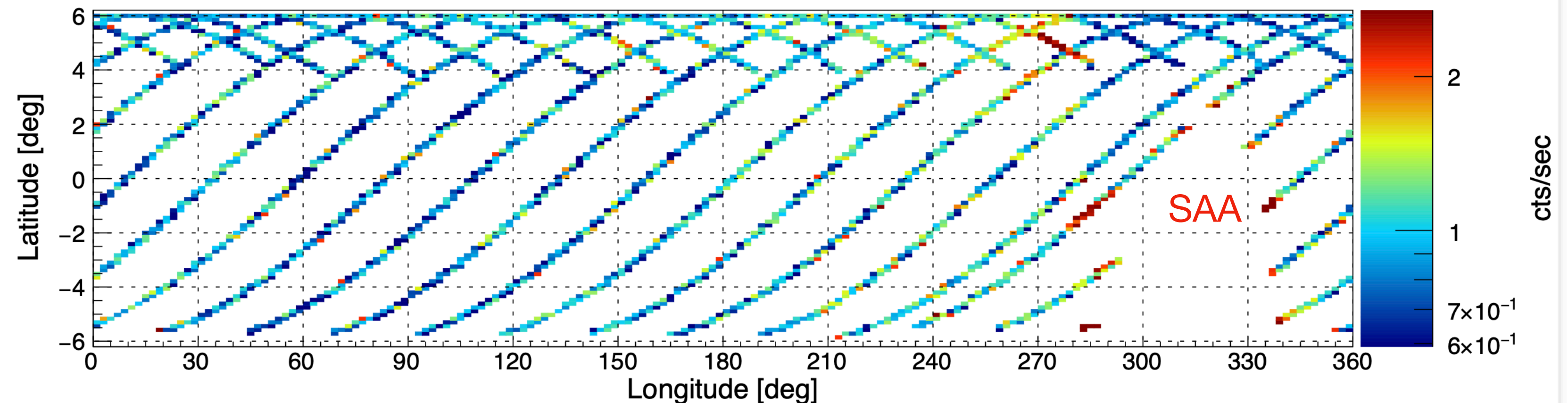
# Mitigation

## What does a “radiation belt” background look like?

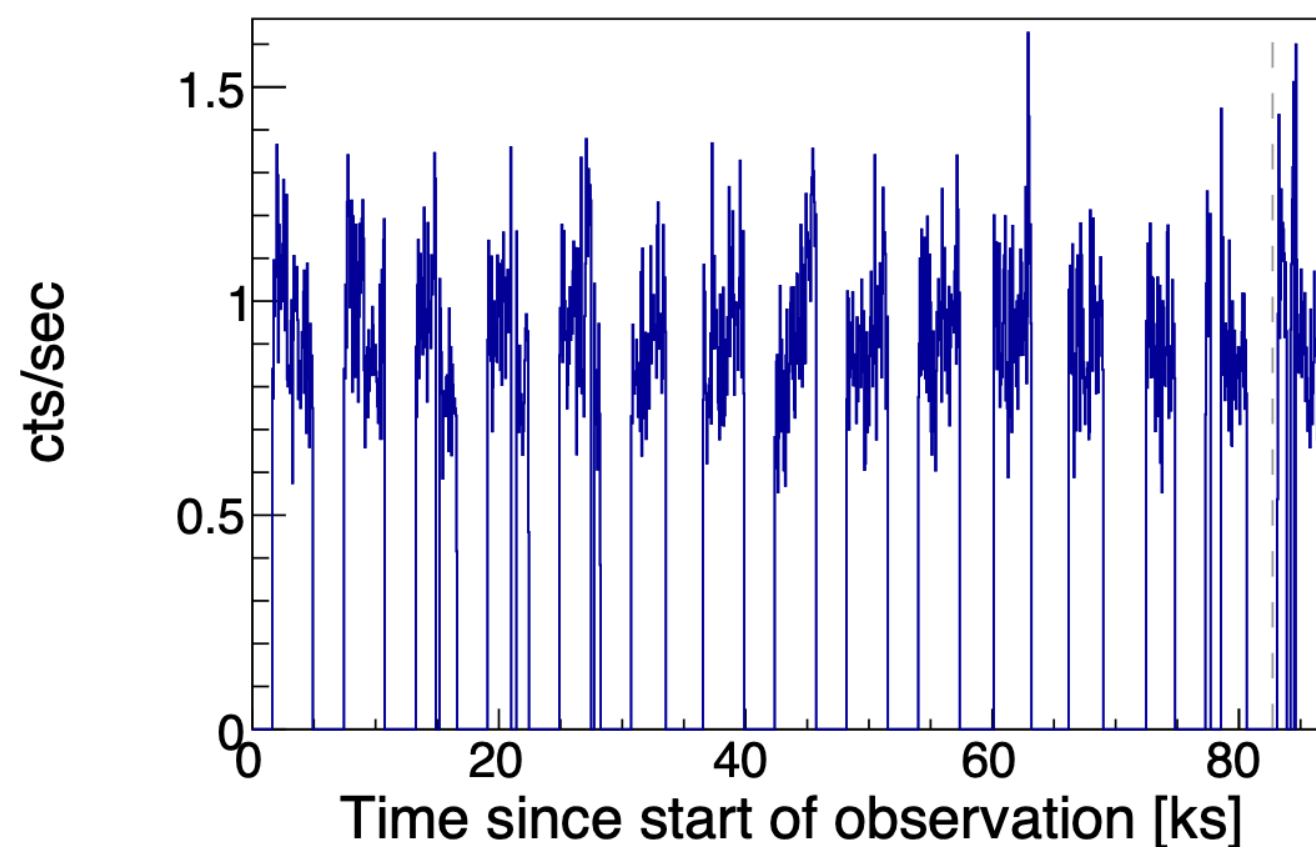
nucalcsaa options: nucalcsaa task not used!



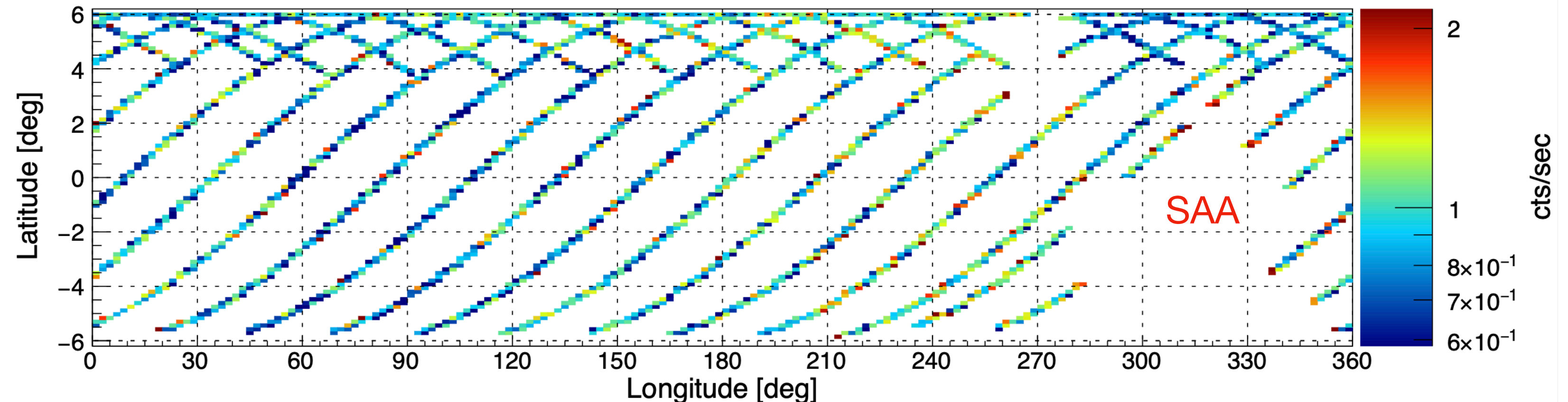
Effective exposure time: 48003 sec (100.0%)



nucalcsaa options: --saacalc=3 --saamode=optimized --tentacle=yes



Effective exposure time: 45691 sec (95.18%)

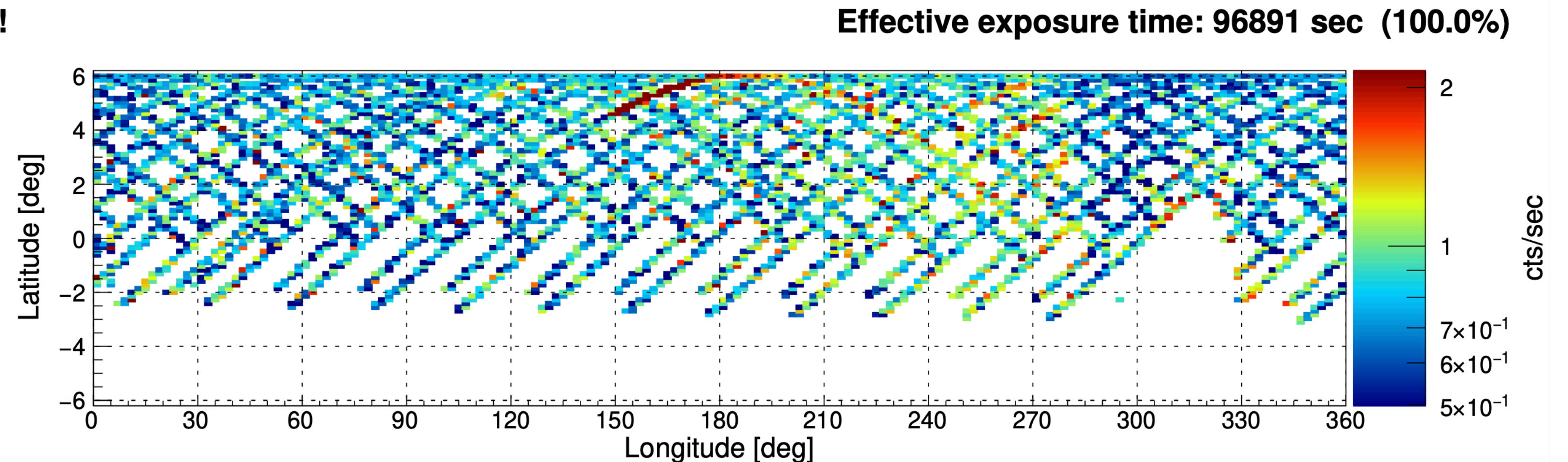
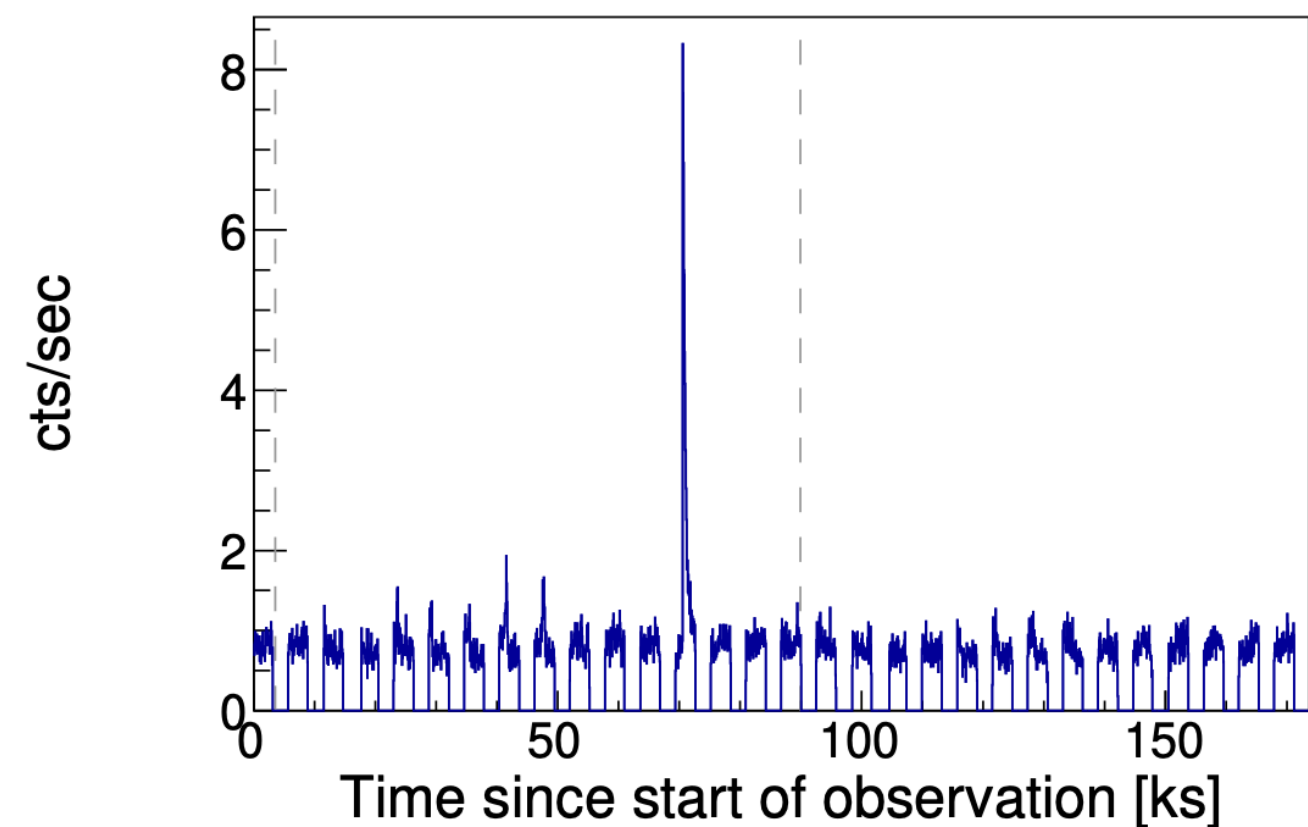




# Mitigation

What does a “solar flare” background look like?

nucalcsaa options: nucalcsaa task not used!



M-class solar flare seen ~78-degrees from the Sun

No standard filters designed to remove these, but easy to identify

Can roll your own GTI to remove these in XSELECT

# Mitigation

## What does “stray light” look like?

Flux from nearby sources (1-3 deg) illuminating FoV

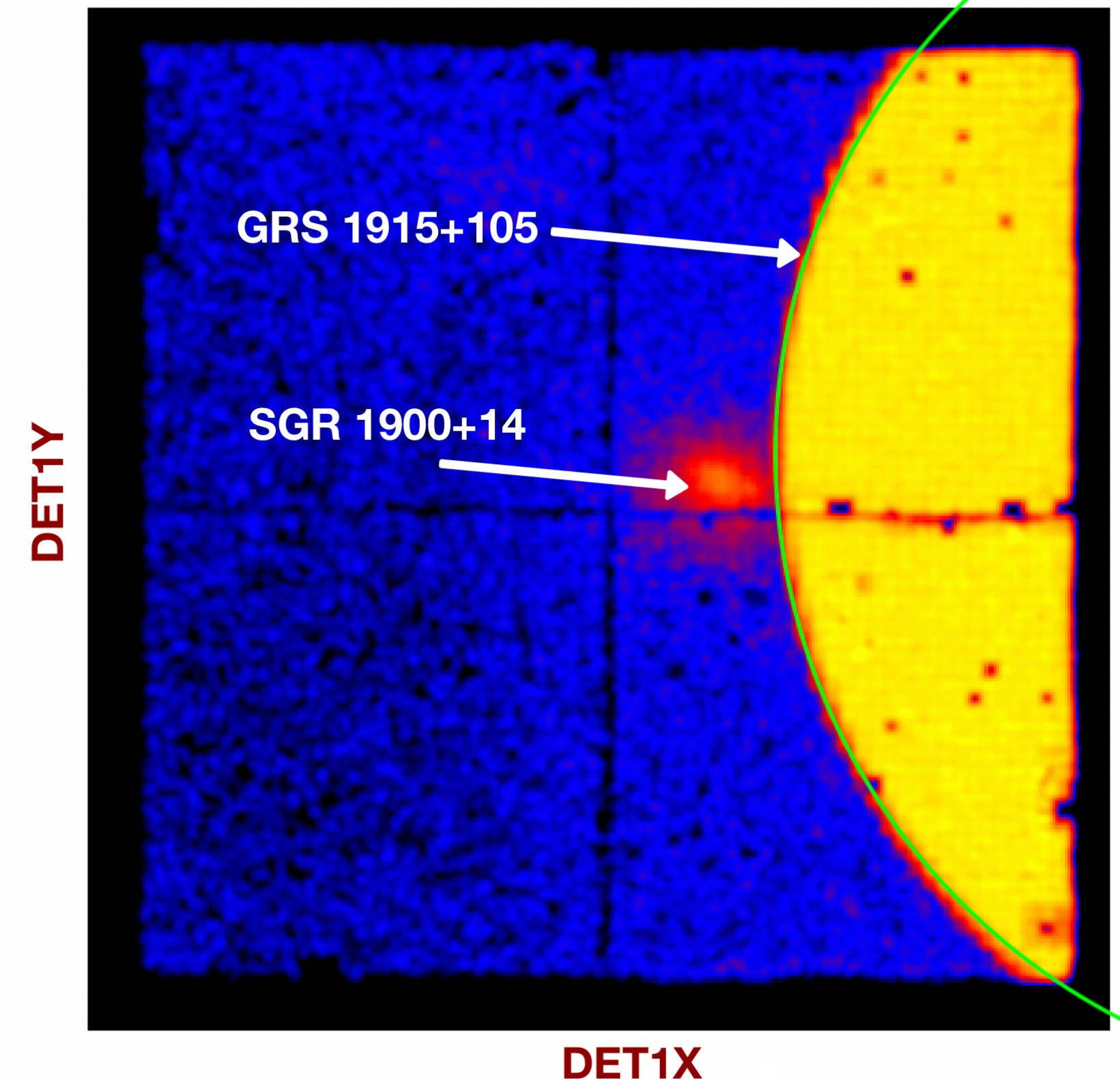
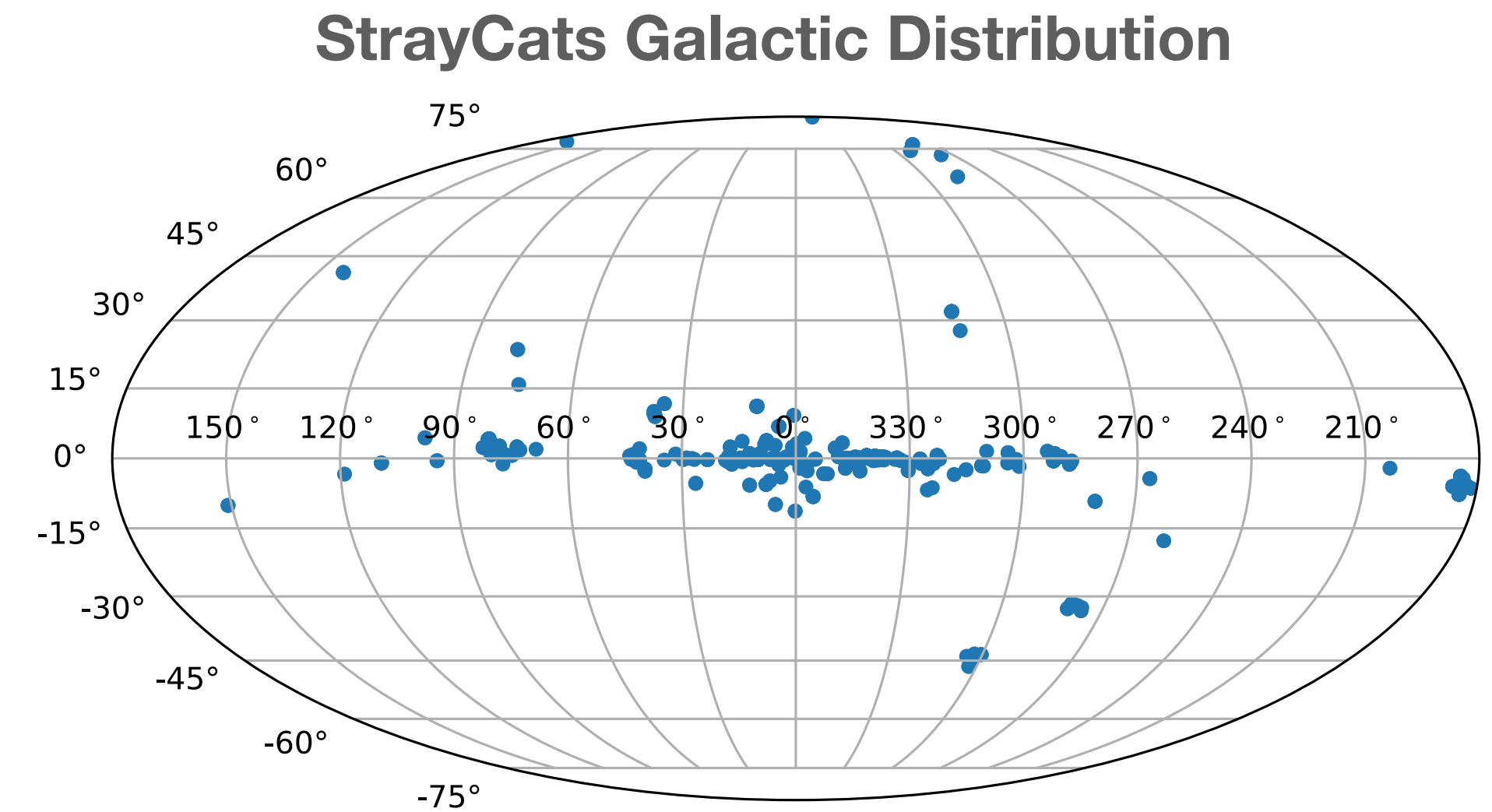
Important for galactic sources

Usually easy to identify, but not always

StrayCats team has checked almost all observations in detail

Check your observation:

<https://nustarstraycats.github.io/straycats/>





# Modeling

**I've done the best I can, but I hear you're not supposed to subtract background....**

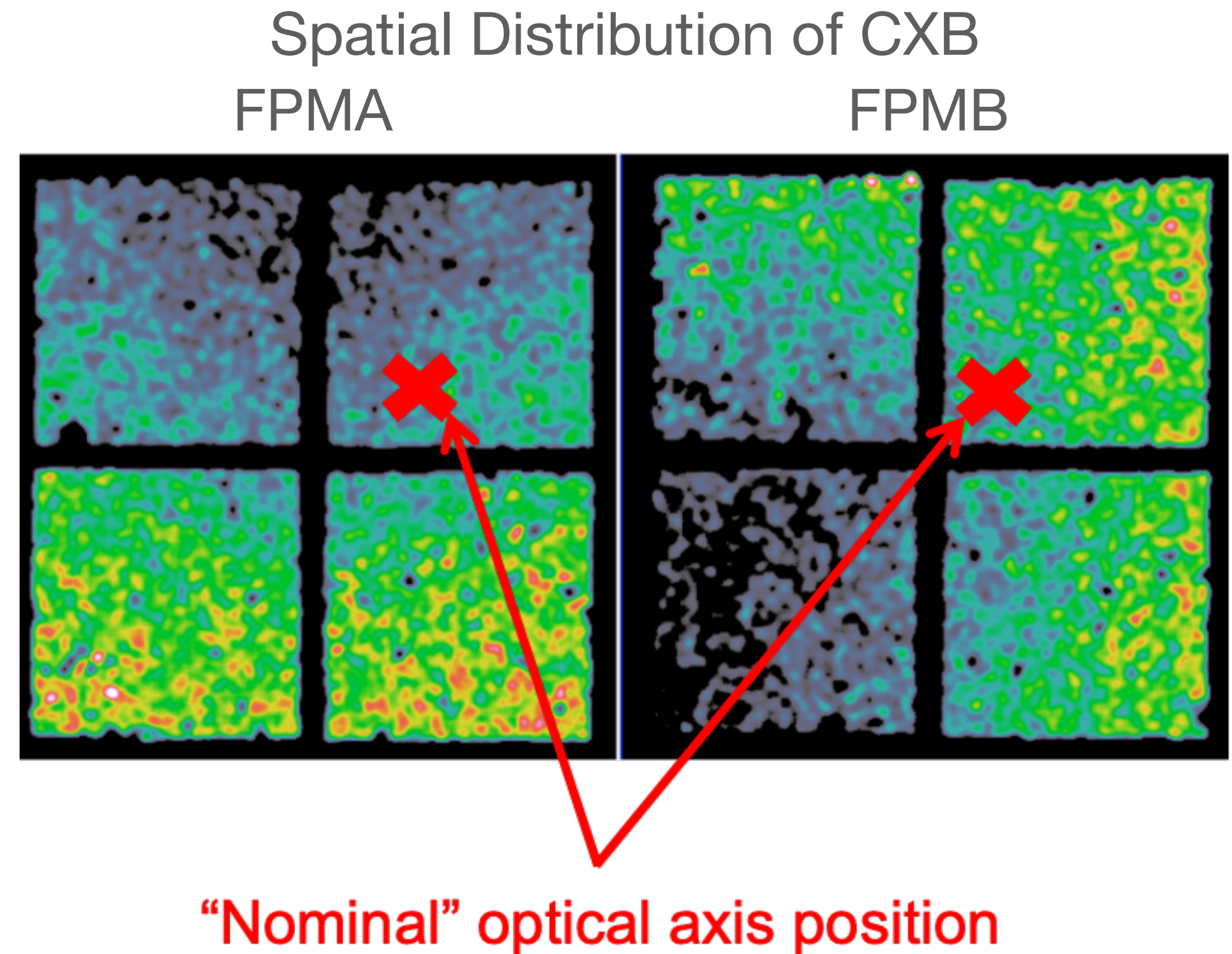
Aperture Cosmic X-ray Background:

Spatial variations present due to geometry

Internal continuum + lines:

Detector-to-detector variations due to CZT thickness

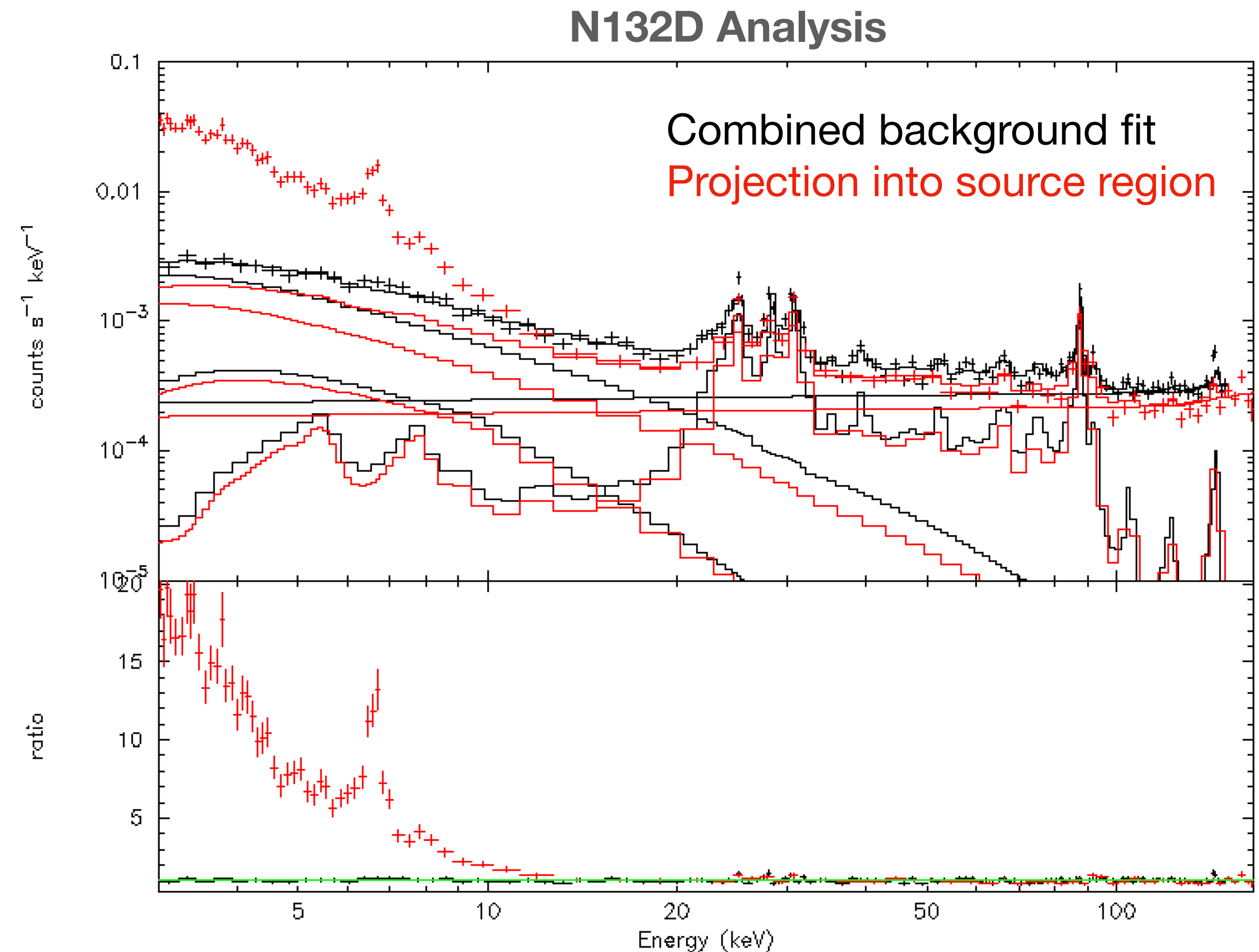
Spatial dependance modeled using nuskybgd





# Modeling nuskybgd

- Latest python implementation and walkthroughs here:
  - <https://github.com/NuSTAR/nuskybgd-py>
  - Original author Dan Wik
  - python port by Qian Wang
  - Further tweaks and examples by BG
- Capabilities:
  - Broadband (3-160 keV) background modeling with appropriate response
  - Produce background models for arbitrary source locations in FoV
  - Produce band-selected background images



# Modeling - Caveats

## What gremlins do you need to know about?

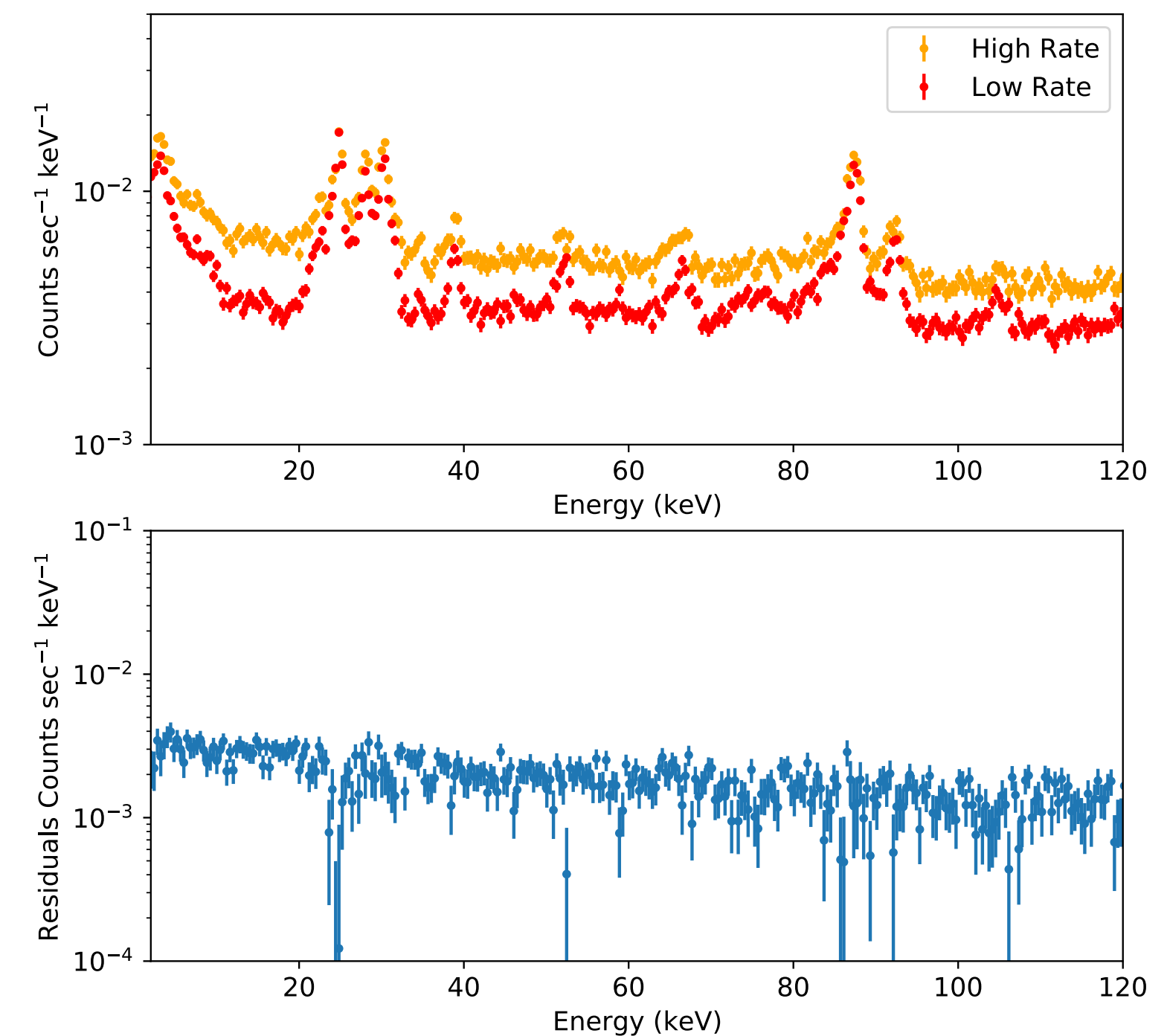
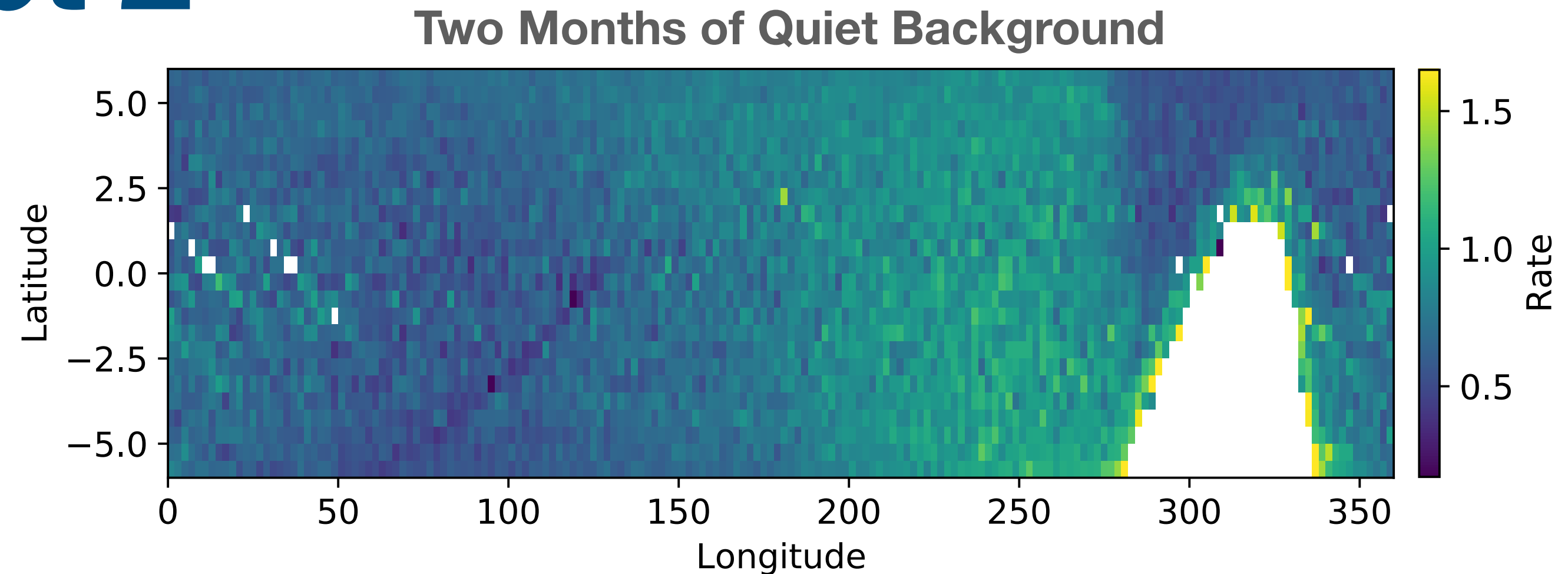
- Solar component currently un-modeled.
  - Mitigate solar flare backgrounds first
    - Worked example shown in nustar-gen-utils
  - For extremely sensitive science\*, use NuSTAR only in Earthshadow
    - \*If you can afford the exposure time loss
- Galactic ridge hard X-ray emission currently un-modeled in python version
  - IDL version has an early spatial template
  - Can be identified by “Fe lines” in background regions within  $\pm 10$  deg of the plane.



# Modeling - Caveats pt 2

## Time dependence

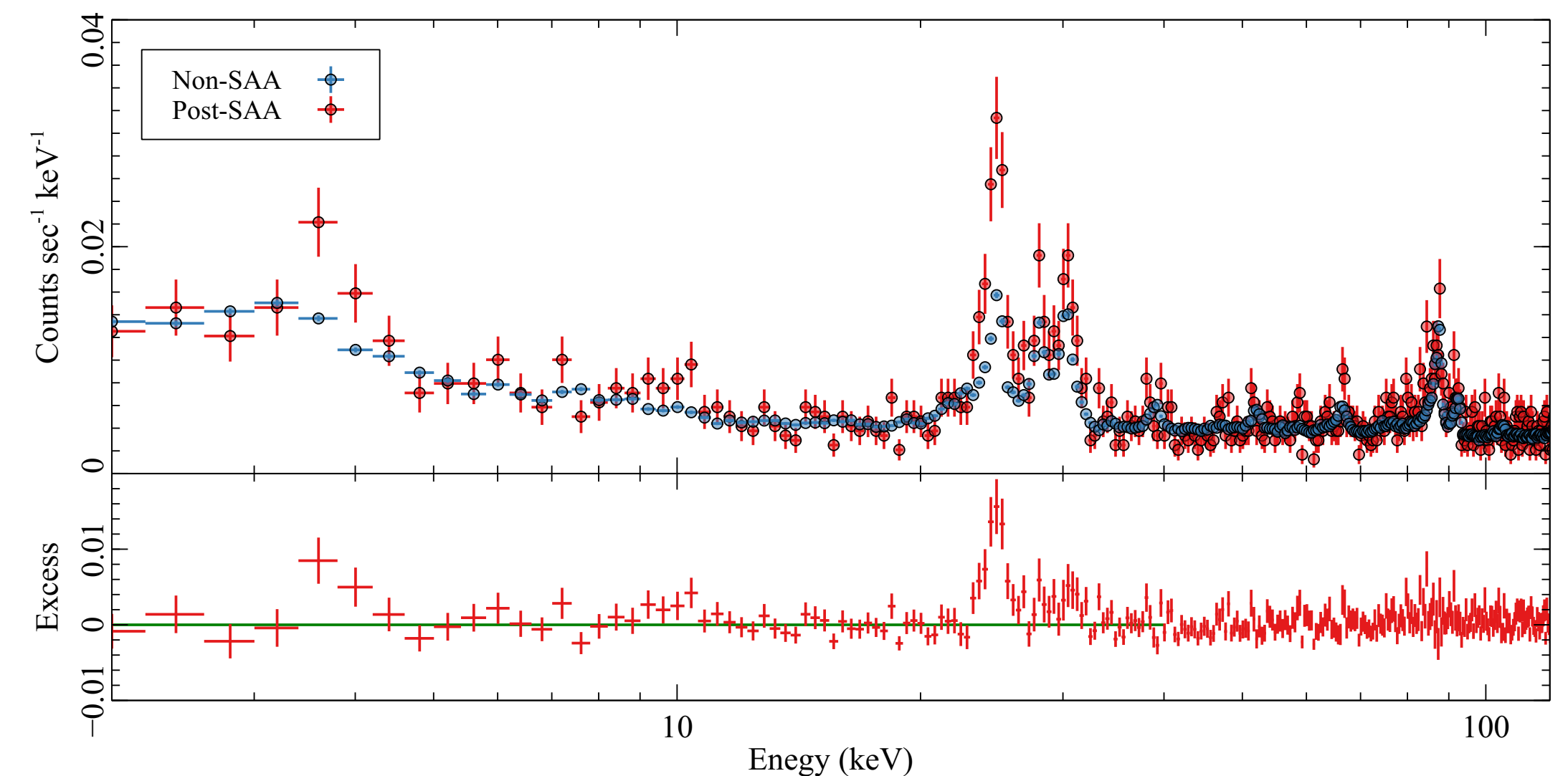
- Continuum has time dependence based on geomagnetic magnetic rigidity cutoff
- Strength will depend on space weather + location in orbit while source was being observed



# Modeling - Caveats pt 2

## Time dependence

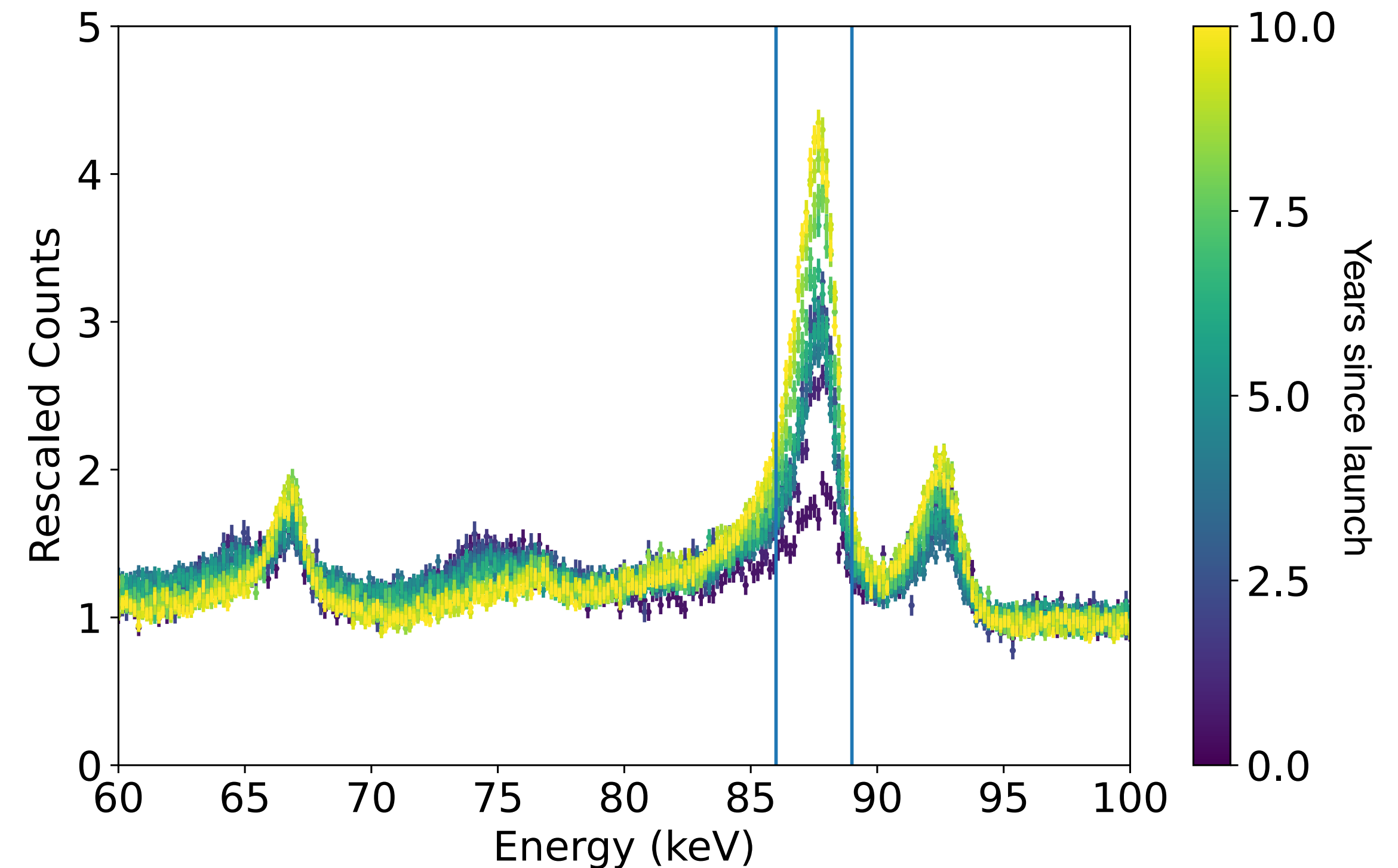
- 24 keV line has SAA-memory
  - Unclear what line this is (lots in this region), but seems to be strongest in ~10 minutes post-SAA
- Strength of line in total spectrum will depend on time spent in post-SAA region



# Modeling - Caveats pt 2

## Time dependence

- 88 keV line-to-continuum ratio has grown over time
  - Half-life is  $\sim 1.25$  years
  - Monotonically (but not linearly) increasing with time
- Relative strength of line will depend on epoch of observation





# Take aways

- Mitigate background first, when possible
  - Background filtering reports for all observations can be available
- Various background components vary over time:
  - Documented in this paper
  - Time-dependence means that “database” of synthetic backgrounds probably not worth producing or useful
- Use nuskybgd to produce synthetic models
  - Need to allow strength of internal continuum, overall normalization of lines, and \*relative\* normalization of 88 keV, and 24 keV lines to vary
  - All demonstrated in worked example on GitHub